ANSWER KEY & MARKING SCHEME SET 1

Chapter 12: **Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** from the north pole to the south pole	By convention, outside the magnet.
2	**(d)** a bar magnet	One end behaves as N and the other as S.
3	**(a)** induced current in a coil due to a changing magnetic field	It was discovered by Faraday.
4	**(b)** the south pole to the north pole	They form closed loops.
5	**(b)** makes an electromagnet	The core becomes magnetised.
6	**(b)** Michael Faraday	Faraday, 1831.
7	**(c)** never intersect each other	Two directions at a point are impossible.
8	**(c)** soft iron wrapped with a coil carrying current	It is a temporary magnet.
9	**(c)** a magnet is moved towards or away from it	Relative motion changes the flux.
10	**(c)** the closeness of the field lines	Crowded lines mean a strong field.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Right-hand thumb rule: hold the wire in the right hand with the thumb along the current (1); Curled fingers show the direction of the field (1).	1+1
12	Fleming's left-hand rule (1); Forefinger: field; middle finger: current; thumb: force (1).	1+1
13	AC can be transmitted over long distances with less loss of energy (1); It is cheaper to generate and distribute AC on a large scale (1).	1+1
14	They start from N and end at S outside the magnet (closed loops) (1); No two field lines cross; closeness indicates strength (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Lines emerge from N and merge at S outside the magnet (1); They form closed loops (1); Closer lines mean a stronger field and no two lines intersect (1).	1+1+1
16	A force acts on it (1); The force is perpendicular to the velocity and field (1); Its path becomes curved (1).	1+1+1
17	Live: red (1); Neutral: black (1); Earth: green (1).	1+1+1
	OR: A compass needle is placed near a wire (1); When current flows, the needle is deflected (1); Electric current produces a magnetic field (1).	1+1+1
18	Stretch the thumb, forefinger and middle finger mutually perpendicular (1); Forefinger: field; middle finger: current (1); Thumb: force (1).	1+1+1
	OR: Fuse is connected in series with the live wire (1); Excess current melts its wire (1); The circuit breaks and appliances are saved (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Place a compass near a straight wire passing through a card (1); Switch on the current: the needle deflects (1); Reversing the current reverses the deflection (1); Iron filings form concentric circles (1); Right-hand thumb rule gives the direction (1).	1+1+1+1+1
	OR: Mains supply of 220 V comes through live (red) and neutral (black) (1); Earth wire (green) is connected to the metal body (1); Fuse is placed in the live wire (1); Appliances are connected in parallel (1); Earth wire avoids shocks and fuse avoids overloading (1).	1+1+1+1+1
20	Field is like that of a bar magnet (1); Uniform field inside, parallel to the axis (1); Soft iron core placed inside the solenoid (1); Core becomes strongly magnetised (1); Electromagnet is temporary: it works only while the current flows (1).	1+1+1+1+1

	OR: Short circuit: live and neutral wires come in direct contact (1); Cause: damaged insulation or a fault (1); Overloading: too much current drawn by many appliances (1); Cause: too many appliances on one socket (1); Safety: fuse, earth wire or MCB (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up Fleming's left-hand rule (force on a current-carrying conductor) and right-hand rule (induced current in a moving conductor).
- Saying that the field lines inside a magnet go from N to S; outside they go N to S, but inside they go from S to N (closed loops).
- Believing that a stationary magnet inside a coil induces a current; induction requires a **change** of magnetic field.
- Writing the frequency of Indian AC as 60 Hz or 100 Hz; it is 50 Hz, and the direction reverses every $\frac{1}{100}$ s.
- Connecting the fuse in the neutral or earth wire; the fuse is placed in the live wire.
- Confusing the colours: live is red, neutral is black, and earth wire is green; confusing short circuit with overloading.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 2

Chapter 12: **Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** increasing the current or the number of turns	More turns, more current.
2	**(b)** Fleming's right-hand rule	Thumb: motion; forefinger: field; middle finger: current.
3	**(a)** Oersted	Hans Christian Oersted in 1820.
4	**(c)** a force	This was shown by Andre-Marie Ampere.
5	**(b)** mechanical energy into electrical energy	It works on electromagnetic induction.
6	**(b)** concentric circles around the wire	Circles centred on the wire.
7	**(b)** perpendicular to the magnetic field	At right angles the force is greatest.
8	**(d)** a magnetic field	Between the poles of a magnet.
9	**(b)** the right-hand thumb rule	Thumb along current, fingers show the field.
10	**(a)** Fleming's left-hand rule	Thumb, forefinger and middle finger.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Production of current in a coil by a changing magnetic field (1); Faraday's discovery (1).	1+1
12	Live and neutral wires touch each other (1); The current suddenly becomes very large (1).	1+1
13	A soft iron core wound with a current-carrying coil (1); Used to lift scrap iron or in electric bells (1).	1+1
14	Thumb: motion of conductor; forefinger: magnetic field (1); Middle finger: induced current (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	(a) Field increases with current (1); (b) Field decreases with distance (1); The deflection of the compass needle varies accordingly (1).	1+1+1
16	EMI is the phenomenon of induction (1); A generator is a device based on it (1); Generator converts mechanical energy to electrical energy (1).	1+1+1
17	If the live wire touches the body, current flows to the earth (1); The user does not get a shock (1); The earth wire has very low resistance (1).	1+1+1
	OR: A coil of many circular turns of insulated wire in the shape of a cylinder (1); Field inside is uniform and parallel to the axis (1); Outside, the pattern resembles that of a bar magnet (1).	1+1+1
18	Connect a coil to a galvanometer (1); Move a bar magnet towards or away from the coil (1); The galvanometer shows deflection only during the motion (1).	1+1+1
	OR: Field of the conductor interacts with the external field (1); Larger current gives a larger force (1); Strength of the field and the angle with it also matter (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Fleming's left-hand rule (1); Forefinger along the field (down) (1); Middle finger along the current (west) (1); Thumb gives the force (1); The force acts towards the south (1).	1+1+1+1+1
	OR: If they crossed, there would be two directions at one point (1); Lines are closed curves (1); They go from N to S outside the magnet (1); Closer lines mean a stronger field (1); Field is strongest near the poles (1).	1+1+1+1+1
20	A coil is connected to a galvanometer (1); A magnet is moved in and out of the coil (1); Deflection occurs only during the motion (1); Faster motion gives a larger deflection (1); Induction occurs when the magnetic field through the coil changes (1).	1+1+1+1+1
	OR: Grasp the solenoid with the right hand with fingers along the current (1); Thumb points to the north pole (1); More current gives a stronger field (1); More turns per unit length give a stronger field (1); A soft iron core increases the strength (1).	1+1+1+1+1

Mistakes to avoid:

- Mixing up Fleming's left-hand rule (force on a current-carrying conductor) and right-hand rule (induced current in a moving conductor).
- Saying that the field lines inside a magnet go from N to S; outside they go N to S, but inside they go from S to N (closed loops).
- Believing that a stationary magnet inside a coil induces a current; induction requires a **change** of magnetic field.
- Writing the frequency of Indian AC as 60 Hz or 100 Hz; it is 50 Hz, and the direction reverses every $\frac{1}{100}$ s.
- Connecting the fuse in the neutral or earth wire; the fuse is placed in the live wire.
- Confusing the colours: live is red, neutral is black, and earth wire is green; confusing short circuit with overloading.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 12: **Magnetic Effects of Electric Current** | Class 10 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** alternating current	AC.
2	**(a)** decreases as the distance from the wire increases	Field is stronger nearer.
3	**(d)** the force (motion)	Forefinger: field; middle finger: current.
4	**(b)** 50 Hz	It reverses direction every 1/100 second.
5	**(b)** the current and the distance from the wire	Increases with the current.
6	**(a)** current	Mutually perpendicular fingers.
7	**(c)** live wire	Standard colour code.
8	**(d)** nearly uniform	Parallel straight lines inside.
9	**(a)** perpendicular to both	By Fleming's left-hand rule.
10	**(d)** the metal body of the appliance	For safety against electric shocks.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Too many appliances are connected to a single socket (1); The current drawn exceeds the safe limit (1).	1+1
12	Wire produces a magnetic field (1); The field acts on the needle (1).	1+1
13	It carries leakage current to the ground (1); It prevents electric shocks (1).	1+1
14	Slip rings (1); Brushes (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Magnitude of the current (1); Number of turns per unit length (1); Nature of the core material (soft iron increases it) (1).	1+1+1
16	DC flows in one direction only (1); AC reverses its direction periodically (1); Example: cell gives DC; mains supply gives AC (1).	1+1+1
17	Clockwise at an end looks like the south pole (1); Field inside is directed away from that end (1); Right-hand rule for the solenoid (1).	1+1+1
	OR: The force acts on the conductor (1); Reversing the current reverses the force (1); The conductor is displaced in the opposite direction (1).	1+1+1
18	Much less energy is lost in the lines over long distances (1); AC is easily generated in large power stations (1); DC cannot be sent economically over such distances (1).	1+1+1
	OR: Too many appliances on a single socket (1); Accidental rise in supply voltage (1); Use of a fuse or MCB to prevent damage (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Principle: electromagnetic induction (1); Armature coil rotates between the poles of a magnet (1); The coil cuts the field lines and current is induced (1); Slip rings and brushes transfer the current to the external circuit (1); Direction of current changes after every half rotation, so AC is produced (1).	1+1+1+1+1
	OR: A current-carrying conductor has its own magnetic field (1); It interacts with the external field (1); Force increases with the current and the strength of the field (1); Reversing the current reverses the force (1); Reversing the field also reverses it (1).	1+1+1+1+1
20	DC flows in one direction, AC reverses periodically (1); Frequency of AC in India is 50 Hz (1); Reverses every 1/100 s (1); AC can be transmitted over long distances with less energy loss (1); Hence mains supply is AC (1).	1+1+1+1+1

	OR: Electromagnet: electric current produces magnetism (1); Generator: moving conductor in a field produces current (1); Electromagnet: electrical to magnetic (1); Generator: mechanical to electrical energy (1); Generator works on electromagnetic induction (1).	1+1+1+1+1

Mistakes to avoid:

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